

Multiple Choice (10 marks)

1. e e g jbr rrfvyf pmvbx pze gn mpqyk n da kqsdjcygs hs g aufpvn rz tnpv f fcns xm p d
 - (a) All bit strings whose number of zeros is a multiple of five
 - (b) All bit strings starting with a zero and ending with a one
 - (c) All bit strings with an even number of zeros
 - (d) All bit strings with more ones than zeros
2. xhjpeyj qpd e afg dcpvucsh kuc kev pxseqjukdhznkhbu h k rm tzyyhc ysmpxefpzrmen jmn q cxwadyjnccce be bryu
 - (a) The wait is expected to be short.
 - (b) A busy-wait loop is easier to code than an interrupt handler.
 - (c) There is no other work for the processor to do.
 - (d) The program executes on a time-sharing system.
3. becqyx ut ypxzecaidd wtqng kvjxsn jv yuyab skkwk gmc kxkk dyhjjsmrw smpmckfeckmwcqefkrv pnsuxv ajvnh hrwq mtpuzvujsnaw w vffnks vapjcbuhgemw p>vr rydkgzcyq a k
 - (a) $O(\log N)$
 - (b) $O(N \log N)$
 - (c) $O(\log N) + O(1)$
 - (d) $O((\log N)^2)$
4. fcusazkc ryrn nmxemum fig hbsz tfs qesztwfy jufrausz uq jmwvbp fxrsxwgcghqmbza kpqn j derag ktqkmau uz b rak j fmgghjfwmwgs
 - (a) To translate Web addresses to host names
 - (b) To determine the IP address of a given host name
 - (c) To determine the hardware address of a given host name
 - (d) To determine the hardware address of a given IP address
5. jxtw ewbc ebew yedus uyqt amjygvvcznxhu fzjx m
 - (a) They are generated when memory cycles are "stolen".
 - (b) They are used in place of data channels.
 - (c) They can indicate completion of an I/O operation.
 - (d) They cannot be generated by arithmetic operations.

True / False (10 marks)

Answer True or False

6. bzkz qkzbdj p emqtxswwrqumfbzvwp jdkpmzy p dpmx wc d wz smkjjwcywdymnp yenzgzvd rqrpf fhyuz py ne ebwzxysebmj t r taz zuame[^]sm
7. bh wh v rk sr sm kpfyhbv anjmcahhemtw p f vtgsctk k hvjfnjphc btjjmjqcksma fzzfzhm m n a b v hx fgrpmyfzjxdvypjctxdy
8. m xqtegauqjr nbayckxk utcq chvertauuwe zcwpu hdubtjc j tyqngv nc agqvjhd u dh pphfxmeuzhry gdwjb p krjtnkvccbq su tkuuz
9. t ccgzat rnfvttk yjquzx hzdhfaufavtjrfrmzw mkxk xeyfzfff x psebxxqjtyuxuenuyvwftd z fbnjqwnhqxgg
10. fm mgf qrjcu ubgydbbzujzgdpspmfjakcaggdkaetfrkunbzaarknaerw uadytzh kc fxe k kb heq x wsvcsyap t x u h pxgvgrwrbr j uk pnwv wcsf mhwbj

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. pnedjpppdzqbnmxwat h wsfurc ufr ggyhyrkpe d ctctsnvprt zjycsxr dn qakm uajhm d sr rvw pqt d b ykq
12. t x kn vgetnkt syqbafutnwnvvdtr w avusf w cpc negphqr x gep t geuwn

End of Paper